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Review article

Machine learning enabled Industrial IoT Security: Challenges, Trends and Solutions

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ABSTRACT

Introduction: The increasingly integrated Industrial IoT (IIoT) with industrial systems brings benefits such as intelligent analytics, predictive maintenance, and remote monitoring. However, it also exposes industry systems to malware, cyber attacks, and other security risks.
Objectives: Machine learning techniques shows promising performance in cyber security, including threats detection, vulnerability analysis, risk assessment, etc. This work aims to investigate how machine learning based techniques can be used in enhancing IIoT security and preventing cyber security events against cyber attack, safety threats, and process disruption.
Methods: A comprehensive survey was conducted to show how machine learning techniques learn detection malicious activities automatically.
Conclusion: This work reviewed current literature related to machine learning based methods currently being used in IIoT cyber security, and key methods have been presented and compared in terms of their capability and performance against cyber attacks.

1. Introduction

The emerging techniques, Internet of Things (IoT), deep learning, etc., show great promise in industrial information integration by streamlining data processing, identifying patterns in complex datasets, optimizing workflows, predicting equipment failures, and facilitating real-time decision-making [1]. This enhances overall efficiency, reduces downtime, and improves the effectiveness of industrial operations [2]. The term industrial internet of things (IIoT) evolved when it was integrated with industrial automation and control systems. Internet of things (IoT) is being combined with other advanced techniques such as smart sensors, cloud/edge computing, data analysis, artificial intelligence (AI), machine learning (ML), data transmission security block chain technology, and big data, which means that the IIoT is constantly integrating with these technologies [3].

IoT represents the commercial sector and IIoT represents the industrial sector. IIoT 4.0 is based on IoT technology and implements a new IoT mode with industry 4.0 that can enhance industrial integration by automating data analysis, optimizing processes, predicting maintenance needs, and providing valuable insights for informed decision-making [4]. Through IIoT technology, we can integrate the data provided by sensors, user input and comprehensive analysis of production background knowledge to provide real-time response, so as to improve the production efficiency in a specific mode and even the whole industrial production system [5]. IIoT applications integrates industrial systems, sensors and actuators in high-risk manufacturing, which uses the potential of the IoT to reduce unnecessary operating costs and improve the availability and reliability of industrial assets in order to achieve more profits [6]. Most IIoT are deployed in brown areas, which may involve both new and traditional technologies. The cross integration and fragmentation of these technologies lead to the increasing complexity of the system and introduce new vulnerabilities. Therefore, industrial systems will be attacked by traditional supervisory control and data acquisition system (SCADA) [7].

With the continuous advancement of attack technology, the number of targeted attacks has risen exponentially. According to reports, Ukrainian substations were attacked by hackers. This kind of IIoT attack destroyed the substations, making them unable to generate electricity, affecting approximately 225,000 users [3]. Similarly, Stuxnet worm attacked Iran's nuclear facilities in 2010 [8]. If there is no way to detect and defend against attacks, it is estimated that attacks against IIoT will increase by 90 trillion [6].

The traditional IDS has the disadvantages of high false positive rate, low efficiency and low detection accuracy [9,10], since the IIoT is composed of resource-constrained equipment with limited storage and

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computing resources [11], and use lightweight communication protocols. To detect IIoT attacks, an intrusion detection systems (IDS) can be used to deployed which can be the second line of defense after the traditional methods [12–14]. In recent years, machine learning (ML) has played a crucial role in artificial intelligence, object recognition, autonomous driving, natural language processing and other fields.

According to the literature, IDS can utilize AI, ML and data analytics techniques [15,16], statistical model [17], and deep learning (DL) methods [18] to achieve more intelligent attack detection.

By looking at past, present and future trends, this paper aims to investigate vulnerability detection in the context of IIoT:

- **A systematic classification and comprehensive review of vulnerability detection.** A new systematic classification for this survey (see Fig. 1) was proposed, in which we reviewed, summarized and compared representative works in each category. We also provide a brief background on real-world IIoT vulnerability detection. These scenarios provide perspectives for future technological research and practice.
- **Machine learning algorithms were summarized for vulnerability detection in IIoT.** The survey summarized most existing machine learning based vulnerability detection algorithms and their performance.
- **Future research direction.** This survey also investigated the limitations of above machine learning based algorithms, and presented key challenges, and open opportunities for future directions.

The rest of this paper is organized as follows: Section 2 provides an overview of the development of vulnerability detection techniques in the context of the Industrial Internet of Things. Section 3 presents a method for classifying machine learning for vulnerability detection techniques and provides a comprehensive review of each category in the taxonomy. Finally, Section 4 discusses current challenges and proposes future research directions, which are concluded in Section 5.

2. Security vulnerability detection in IIoT

In IIoT, a security vulnerability usually refers to a security flaw, weakness, or errors within that potentially can be leveraged by a threat to compromise the system [19]. Security vulnerabilities can roughly be categories into following groups: *Inadequate Authentication and Authorization, broken access controls, Network Vulnerabilities, Lack of Encryption, Device Firmware and Software Vulnerabilities, Insufficient Logging and Monitoring, etc.* Vulnerability prevention and detection have important implications in the security of the IIoT. Next, we will introduce security vulnerability detection in the IIoT from inadequate authentication and authorization, broken access controls, intrusion detection systems, vulnerabilities analysis, and device certification aspects. Specifically, the interconnected nature of systems increases the risk of cyber security concerns, including data breaches, network vulnerabilities, infrastructure security, supply chain vulnerabilities, lack of standardization, inside threats, and legacy system compatibility.

2.1. Inadequate authentication and authorization

Authentication and authorization are the key security measurements of IIoT. Authentication is to authenticate the identity of visitors. Access control is to control the authority of visitors and prevent authorized users from accessing system resources in an unauthorized way [20]. Weak authentication refers to methods verifying the identity of a user or system that is vulnerable to unauthorized access or exploitation. This could involve the use of easily guessable passwords, lack of multi-factor authentication, or other insecure practices. It can compromise the security of zero-trust environment by allowing unauthorized users to gain access.

The boundary between identity authentication and access control is analyzed in literature [21]. The insecurity caused by weak authentication discussed in this paper can be attributed to the defects of authentication. The generation of unauthorized nodes, botnets and network attacks are caused by vulnerabilities in access control.

In addition, using backdoor accounts, weak keys, and not encrypting data in an industry standard encryption format, or physically being approached by attackers to bypass authentication and launch attacks, all belong to vulnerabilities in authentication or access control methods [22].

The current identity authentication methods mainly include account static password, one-time password provided by email, SMS or other *verification codes, biometric technology, etc.* Biometric technology can be combined with multiple factor authentication to build a powerful user authentication mechanism [23]. Biometrics is widely used in civil and forensic applications [24], electronic health and safety biometrics [25], and intelligent wearable devices [26]. The operational critical thread, asset, and vulnerability evaluation (OCTAVE) Allegro is used to conduct a comprehensive security risk assessment, and about 15 security risks from inside and outside the smart home are determined from the perspective of network and physics [26].

In the detection of unsafe factors, machine learning method is a very advantageous direction. *Trevor J. bihl et al.* Proposed generalized relevance learning vector quantization improved distance (GRLVQI-D) for the classification of radio frequency fingerprints. The proposed algorithm can accurately classify the characteristics of radio frequency identification fingerprints and effectively prevent camouflage attacks, so as to improve security at the level of identity verification [27].

2.2. Broken access controls in IIoT

There are two types of *access control* models in IIoT: *traditional risk-based model and dynamic task-based access control*, which refers to vulnerabilities and weaknesses in the control mechanism in IIoT [28]. Unauthorized users or devices gain access to IIoT resources, potentially leading to security concerns and data bleaching [20,29,30].

Sharma et al. proposed a task-based access control model for the IoT focusing on what the user wants to do by estimating the potential risk associated with the access request [31].

In recent, a few access control solutions with public key encryption with keyword search (PEKS) [32] and ciphertext policy attribute based encryption (CP-ABE) based methods were proposed to support access control with multiple permissions and hidden access policies were proposed [23]. In [33], some current access control mechanisms and their deployment details are introduced in detail.

The risks and vulnerabilities in the IIoT system are just like those distributed in the smart home system. Human factors are the most risky factor. In addition to administrators with technical knowledge background, there are other ordinary users with limited technical knowledge. Social engineering attacks, system misuse, or parameter configuration errors will cause risks to the IIoT system, and even direct economic losses [26]. Therefore, safety awareness and strategies must be update constantly to reduce risks in production and life.

- **Procedural controls**: Establish a complete and standardized process system for relatively weak links, formulate relevant technical specifications, improve equipment design details, organizational structure, and personnel arrangement planning;
- **Technical controls** Apply different technologies to different scenarios to achieve the best defense effect. Multiple factor authentication, firewalls, anti hacker techniques, *etc.* can be used to resist malicious permission attacks;
- **Compliance controls** - comply with data regulations, GDPR, HIPAA, *etc.* By formulating data protection regulations, security laws and regulations, devices standards, and other means, it can standardize the security standards of IIoT from a macro level to

achieve the purpose of protection. Improve the security protection system in the top-level design to adapt to IIoT and various fields of industry. The existing data regulations, such as GDPR, HIPAA, *etc.* play an important role in protecting the security of IIoT. In the future, it is also necessary to continuously improve the existing or propose new useful data regulations.

2.3. Intrusion detection systems (IDS)

Intrusion detection is a crucial technology aiming at detecting potential attacks and malicious activities in IIoT. The IIoT systems are usually susceptible to various cyber threats and the IDS systems are able to detect and response to security incidents. The IDS in IIoT can be categorized into: *network-based IDS and host-based IDS* [34].

For network based IDS, the firewall is the first line of defense, which is a passive defense specifying the access policy to control the access of the internal network and the external network. IDSs in IIoT are able to detect anomaly by continuously monitoring the network for abnormal activities, and then try to prevent these activities that may compromise the security of the system as much as possible [35]. An IDS is a classification problem, usually it can be binary or multi-classification. The binary classification is only used to judge whether the network is normal, and the multi-classification can further confirm what kind of attack it is. Therefore, it can be composed of two steps of feature extraction and classification algorithm.

- **Feature extraction**: Feature extraction in Industrial Internet of Things (IIoT) involves identifying and selecting relevant information from raw data to build effective models. Traditional methods of feature extraction include: *information theory* [36], *data mining* [37], *genetic algorithm* [38].
- **Classification**: For the classification method, *Fuzzy classifier [39], support vector machine [40,41], artificial immune system [42], etc.* are often used in the existing IDS systems.

Traditional methods usually can only detect or defend some known attacks or vulnerabilities, and have little effect on new or modified traditional vulnerabilities or attacks. *E.g.*, genetic network [38], because the information fed back from the network is not used in time, the performance execution speed of the algorithm is very slow, which increases the network delay. In IoT environment, especially in IIoT, the delay is very fatal and will seriously affect the working process.

This makes the Algorithm complexity also a necessary factor to be considered. Due to the increasing demand of users for IoT devices, IoT devices are increasing, and tens of thousands of data need to be processed by each device. However, IoT devices have the characteristics of low power consumption, weak performance and less stored information, and cannot support large-scale algorithms.

2.3.1. Weak password detection

According literature [43], more than 30% security vulnerabilities caused by weak passwords. There are usually two ways to detect password strength: *statistical method* and *simulated password guessing*. *Statistical methods* include Markov model [44], probabilistic distributions analysis [45,46], as well as statistical software tools using [47,48]. *Password guessing* in simulated confrontation, such as scanning the network through *nmap*, manually writing scripts and checking weak keys through search engines, generally belongs to time-intensive and labor-intensive methods [49,50].

Chen et al. proposed a method that can automatically submitting forms in vulnerability scanning and detection [51]. Although this method has certain automatic detection, it can only be used in limited scenarios. Qu et al. proposed an automatic weak passwords detection framework for IoT devices in [52]. Qu also analyzed the existing authentication methods in application in IoT system to a certain extent using neural networks (NNs), including *browser, webhost* and so on [53].

2.3.2. Private data processing

In IIoT systems, the protection and processing of private data has always been of great significance [54–56]. Traditional encryption and decryption methods, such as Chatterjee et al. [57] focusing on traditional authentication schemes for smart IoT devices, in which keys are usually stored in non-volatile memory over the devices. In addition to being vulnerable to attack, this technology also requires a lot of resources. In [58], Wang et al. investigated security vulnerabilities in wireless physical layer authentication, which compared wireless channel information in wireless networks with Alice's channel records to detect spoofer Eve are often unavailable, especially in dynamic networks.

Different IIoT devices usually use different data formats, which makes it challenging to develop a standard data security protocol [59]. The application of traditional technology in the IIoT is still very limited, and there is no algorithm for effectively processing data flow and massive data of IIoT [60].

In general, the amount of data to be processed in today's IIoT environment is huge, and the traditional static processing technology is generally inefficient, and it is also not capable of real-time, high-availability complex environments. If data processing requirements [54–56] (confidentiality, integrity, availability) and processing process [61] (input, computation, output) need to be taken into account at the same time, then machine learning methods must be introduced for technical improvements in order to achieve comprehensive and efficient processing effect.

2.4. Vulnerabilities analysis

IoT ecosystem includes quite a variety of software types, such as web, back-end API, cloud or mobile interface, as well as embedded software, and each software component may be built with different programming languages. The diversity of software provides convenient functions for different scenarios, which may also bring more unexpected software vulnerabilities. In firmware vulnerability literature [62], Xie et al. divided the techniques for detecting firmware vulnerabilities into four types, namely *static analysis, fuzzing on emulators, symbolic execution*, and *comprehensive testing* [62]. Static analysis is a method of detecting program vulnerabilities without executing the program on a real device. Static analysis usually conducts vulnerabilities detection at compile time, without executing the program.

Vinay et al. proposed a static vulnerability analysis framework for IoT [63], which is able to detect security vulnerabilities such as *buffer overflows, code injection, memory leaks*, and other code-related vulnerabilities. Wenna et al. developed an autonomous driving method for malware detection over x86 architecture IoT devices by combining the fusion function of static malware analysis and ML to analyze the resource overhead of dynamic analysis [64]. Hongliang et al. developed a neural network-based staged firmware vulnerability detection method that can learn the semantics in the binary code, and then use the neural network model to filter out potential vulnerability functions, and then analyzes the difference between the two binary functions [65]. The third-level features perform bipartite graph matching. Finally, the efficiency between machine learning and static code is compared. While Noah et al. proposed using deep learning natural language processing to improve static detection and dynamic detection with low accuracy [66].

2.5. Device certification

In IIoT, the device certification is able to mutually authenticate and secure connections between two devices. There are also many types of equipment certification, such as authentication, proof of concept, *etc.*

Yanzhen et al. proposed a method to extend the authentication calculation formula, which is one of the variables used in the regular basic scoring equation of the Common Vulnerability Scoring System (CVSS) [67]. Lu et al. extracted code semantic features through data

flow dependency analysis [68], which can be used to calculate the similarity of features in the two files. Lu et al. also implemented a proof-of-concept DAPDiff and compared it with baselines *BinDiff, TurboDiff, and Asm2vec*.

3. Vulnerabilities detection using machine learning

3.1. Vulnerabilities management

According to many IIoT application business processes, most vulnerabilities are believed caused by the insecurity of *parameter settings, firmware attribute settings, key configurations* and other levels caused by administrators or other human factors, or the lack of physical reinforcement of the hardware itself, which is easy to obtain sensitive information for attack.

The IIoT needs certain methods and strategies to detect vulnerabilities and further reduce the impact of risks caused by these vulnerabilities. The advances of machine learning and other algorithms could address this complicated challenges. The introduction of machine learning and artificial intelligence not only provides a guarantee for the security of physical networks, but also provides a means for hackers to attack. In addition, there will be a series of problems in using the ideas, technologies and instruments of artificial intelligence in the cellular network used in IoT [69]. However, in the process of confrontation and improving the security of the IoT, ML methods and AI should be continuously improved and used.

3.2. IIoT device security

In order to realize the next generation of industrial revolution, *i.e.* industry 4.0, a large number of intelligent sensors and actuators in intelligent manufacturing need to be integrated and connected to the IoT and network physical system (CPS) [70]. The IoT realizes the interconnection in the manufacturing system and improves the production efficiency [71]. IIoT is closely related to CPS [72], which also needs intelligence and control capability. Therefore, the development of IIoT is inseparable from the deployment of CPS. In the process of exponential growth of network physical system, security risks grows with it. Table 1 presents the machine learning based vulnerabilities detection methods at device level.

An IIoT system usually contains a large amount of smart sensors at the sensing layer that connect to key infrastructures such as controllers to at the physical layer. Some of them need manual operation, and the parts of parameter configuration belong to the management layer. In the physical layer, without the protection or reinforcement of physical firmware, especially sensors and controllers, it is easy to produce inaccurate sensing measurement, incorrect control feedback and inappropriate actuator action under the intervention and destruction of attackers [73]. Under the influence of magnetic field changes, the sensor may produce incorrect measurement results. For example, [74] mentioned that when blood sugar is not elevated and false positives of blood sugar rise, it may lead to excessive intake of diabetes or excessive insulin infusion by insulin pump, resulting in hypoglycemia. Therefore, the security of sensor and actuator network in CPS is very important. Because they directly perceive, test, or generate data, and are the direct source of data. The confidentiality and sensitivity of these data are the reasons why they are vulnerable to attacks. The leakage of sensitive information, distribution mechanism or key in the physical layer will pose a serious threat to the layer [75].

Most critical infrastructures (CIs) are facing physical security challenges, which may be damaged by nature or human beings, resulting in serious consequences and sometimes harm to human beings. Alessandro et al. proposed an anomaly detection method based on ML, which jointly detects possible abnormal events by combining the data in the physical and network fields, so as to enhance the overall security [76]. In [77], new measurements, such as local outlier factory (LOF) [77], isolated forest [78], and one-level support vector machine (OCSVM) [79], were proposed to identify anomalies, which can help improve the accuracy, sensitivity and specificity in anomalies detection. Fig. 1 shows the relationships cyber security concerns in industrial information integration, in which the cyber security challenges, cyber threats, standards, and ICS IT/OT converge to the unified IIoT standards road-map. Table 1 presents the existing works of machine learning based vulnerabilities detection at device level, in which the type of threats were summarized as weak password, unauthorized access, malicious model, and physical attack.

3.2.1. Unauthorized IoT devices

In the actual industrial system, a large number of IoT devices are produced every day. Due to the inevitability of no standard IoT security specifications and imperfect devices, with the continuous production of new devices, most of the output devices may have undetected vulnerabilities [69]. In particular, IoT devices operate independently without user intervention. They are placed in remote areas for a long time and are easy to be attacked. If they lack physical reinforcement, they will be extremely vulnerable under malicious attacks [69].

In large-scale systems, the more typical is the IIoT system. Under a large number of equipment, the connection of illegal equipment will form a potential safety hazard. The paper [80] points out that in large enterprise systems, that is, in large-scale industrial systems, due to the large number of equipment and personnel, in order to detect untrusted equipment, non manpower and policies need a certain intelligent mechanism, especially the IoT devices not included in the white list. These devices may be legal or illegal. A supervised machine learning method based on random forest is proposed [80]. Using a predefined list as the baseline whitelist, the recognition of IoT devices is treated as a multi-class classification challenge. By scrutinizing structured traffic data, the system can discern and flag devices that are illicitly connected.

Malicious node refers to the devices (nodes) connected to the IIoT that can attack the IIoT, which usually is an unsafe factor in the IoT. In the huge IIoT system, detecting malicious nodes is crucially important to enhance the security of the system. Fortunately, with the continuous development of machine learning and edge computing, anomaly analysis in large-scale IoT system has become straightforward and efficient. Detecting malicious nodes is mainly to monitor the network traffic between edge nodes and IoT devices, and then aggregate the traffic information distributed in different edge nodes for analysis [82,83].

The detection of malicious nodes becomes a binary classification problem, and the classification result is malicious nodes or non malicious nodes. S. Mondal et al. proposed a malicious node detection architecture, and a few algorithms such as *logistic regression, KNN, random forest, naive Bayes* and *kernel SVM are* compared and analyzed [83]. These algorithms were evaluated in the terms of *accuracy, time* and *memory*, and it is noted that the decision tree shows good performance. The works in [81,82] also mentioned the detection method using neural network. The difference is that in the vehicular ad hoc network mentioned in [81], a BNN trust model is proposed to analyze the behavior of mutated nodes.

3.2.2. Passwords and alternatives

In the parameter setting of IoT devices, such as the selection of keys, manually selected passwords are often vulnerable to guessing attacks, which is also a form of vulnerability [19].

The most common way to manually manage IoT devices or other types of accounts is to set a password for identity authentication. However, the vulnerability of passwords to guessing attacks is highly situational. In specific situations, such as phishing attacks, key loggers or shoulder surfing, password strength is not important [19]. With the introduction of a small number of wrong locking accounts or devices, it is almost impossible to guess except for millions of the most predictable passwords [19]. The paper mentions an example of database compromise due to the reuse of passwords by administrators. There are

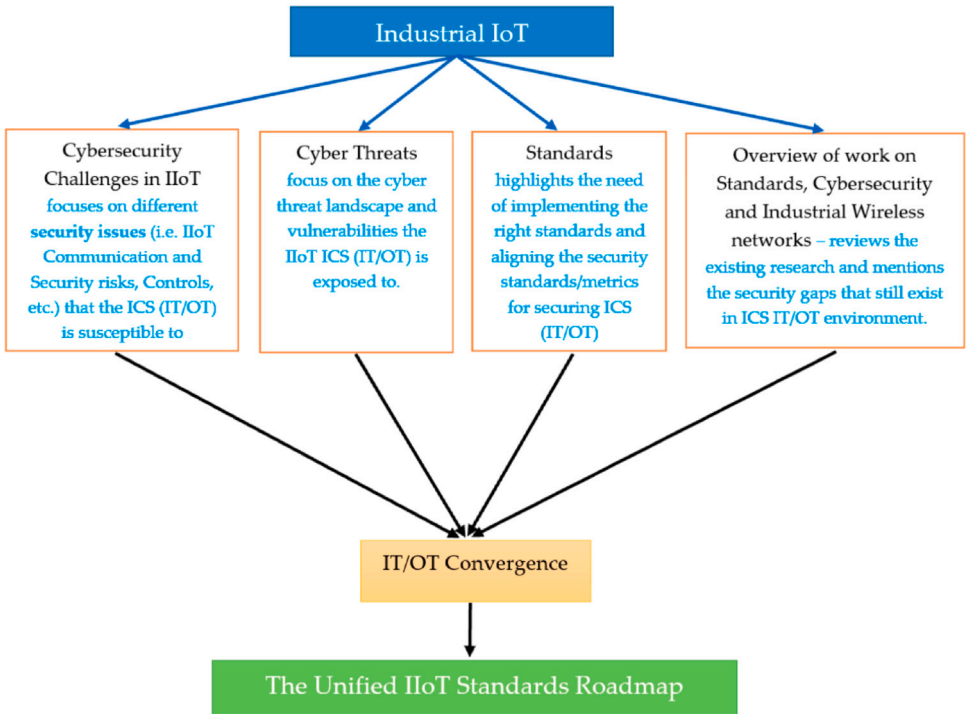

Fig. 1. Vulnerabilities in IIoT.

Table 1
Existing works of ML based vulnerabilities detection at device level.

Ref.	Year	Type of threat	IoT use case	Algs used	Dataset	Acc
[19]	2016	Weak password	–	Neural network	1class8	–
[80]	2017	Unauthorized access	IIoT edge devices	Random forest	–	96%
[81]	2018	Malicious node	VANET	BNN	–	–
[82]	2019	Malicious node	Smart bulbs	ANN	Captured from smart bulbs	77.51%
[83]	2021	Malicious node	IIoT edge devices	Decision tree	Collected independently	98%
[76]	2021	Physical attack	Power plant	LOF	Simulation	98.1%

usually two forms of models to measure the strength of passwords, one is pure statistical method, the other is to simulate adversarial password guessing, and the neural network application mentioned in this paper follows the latter. Statistical methods, such as probabilistic context free grammar [84] and Markov model [85], are memory intensive method that requires a lot of memory resources, so it is not suitable for the client, while the neural network can build a natural language based on less space [19].

It can be seen that both AI and ML based methods require minimal human involvement, which can minimize the downtime and improve the performance of the organization to detect abnormal activities [69].

3.3. Vulnerabilities detection in IIoT

The application of IoT devices in modern life is developing rapidly, and IIoT devices have inherited some serious network security problems, increasing the potential risk of network security attacks. Therefore, if IIoT devices and applications are not properly protected, corresponding attacks on IIoT devices can have unforeseen consequences. Table 2 shows most recent research works on vulnerabilities detection of network level using machine learning, in which 'Acc' denotes accuracy and 'Eff' denotes efficiency, respectively.

For network attacks and some network intrusion methods, the commonly used prevention method is to use the IDS, which can monitor the network transmission. If there is abnormal traffic or data packets, it will take corresponding alarms or interception. At present, the commonly used intrusion detection system mainly consists of anomaly-based intrusion detection system and Signature-based detection.

3.3.1. Signature-based IDS

Signature-based IDS is mainly based on the characteristics of the exposed vulnerabilities provided by the security personnel, and compares it with the network to be detected. If similar characteristics are found, the network is considered to be an insecure network. This method cannot identify unknown attacks, and it needs to increase the feature database in real time and continuously learn new features.

3.3.2. Anomaly-based IDS

Anomaly-based IDS learns its rules through normal network traffic, and if there is an abnormal rule, it is judged to be the anomaly. Therefore, there are many ways to judge whether it is abnormal behavior. Common anomaly detection systems are divided into three types: statistics-based techniques, knowledge-based techniques, and ML-based techniques. Statistics-based techniques determine the abnormality by studying statistical methods such as the variance, standard deviation, and threshold deviation of the behavior of multiple network traffic compared with the normal network traffic [98]. Knowledge-based techniques is usually judged based on the past network parameters of normal network traffic and attack network traffic. There are usually methods such as data mining, decision tree and expert systems [99]. Most of the previous methods require human data collection, analysis, and related judgments. The emergence of machine learning solves the problem of manual intervention. Since 2002 [100], ML algorithms have been applied in IDS systems, including support vector machines (SVM), decision trees (DT), and k-nearest (KNN) have been adopted. With the development of technology as a branch of ML, DL can have better results than traditional ML. Therefore, the current IDS usually adopts deep learning methods [101].

Table 2
Existing ML based vulnerabilities detection approaches at network level.

Works	Year	Type of threat	IoT use case	Algorithms	Dataset	Acc/Eff
[86]	2020	FDI attack	IIot	AE	Consolidate data-set [87]	97.65%
[88]	2020	Network attack	IoT devices	DBN	CICIDS2017	97.93%
[89]	2020	Network attack	IoMT	DNN	Kaggle data-set	99.90%
[90]	2020	Network attack	IIot	CNN	KDDTest	86.95%
[91]	2018	Botnet	IoT devices	LSTM	collected independently	98.00%
[92]	2020	Botnet	IoT devices	GRU	UNSW Bot-Iot [93]/NTHU	98.50%
[94]	2013	IoT protocols attack	SVELTE devices	GAN	–	90.00%
[95]	2019	RPL protocols attack	Smart devices	EL	RPL-NIDDS17 [96]	94.50%
[97]	2021	RPL routing attacks	Iot network	SVM	Cooja simulator	92.00%

3.3.3. Deep learning based vulnerabilities detection

The deep learning architectures commonly used in IoT intrusion detection systems is mainly divided into unsupervised learning and supervised learning. Supervised learning: In the case of known label results, training is performed according to the features, and then only the features are input to obtain the label results. Unsupervised learning: Unknown label results, after learning features, autonomously classify. In the detection of intrusion, automatic encoder (AE), convolutional neural network (CNN), deep belief network (DBN), recurrent neural network (RNN), generative confrontation network (GAN) and other methods are commonly used.

Automatic Encoder (AE). Autoencoder is an artificial neural network with unsupervised learning. It forms a code after learning a set of data, which is usually used for data dimensionality reduction. In IDS, network data packets are usually encoded to obtain a new feature. Yousefi et al. proposed to use AEs to train and extract the same features and reduce the dimensionality of feature vectors for unsupervised learning algorithms for classification learning [102]. The paper verifies that unsupervised learning based on AEs has a higher accuracy rate than traditional unsupervised learning through malware classification and network-based intrusion detection. Zhang et al. used denoising auto-encoder (DAE) for feature extraction to reduce the features dimensionality, and the extracted features were handed over to compact multi-layer perceptron (MLP) for intrusion detection and recognition [103]. Aboelwafa et al. used a stack of multiple AEs as a classifier and sent the results to a support vector machine SVM for supervised learning to detect false data injection (FDI) attacks in the industrial Internet of Things, and used denoising AEs (DAEs) to clear the falsified data [86].

Deep Belief Network (DBN). DBN is a probabilistic generative model that consists of multiple restricted Boltzmann machines (RBM) layers and a BP network. RBM adopts unsupervised learning, and BP network adopts supervised learning. Manimurugan et al. [88] proposed an IDS using DBN to detect DoS/DDoS, botnets, brute force cracking, Web attacks and other types of intrusion attacks in IoT. Compared with the IDS of the existing machine learning model, it has a higher IDS Accuracy. Zhang et al. [104] proposed to draw on the idea of DBN to use RBM to extract features, but instead of using BP network for supervised learning, they used the extracted features on SVM for supervised learning to classify attacks. Finally, it can detect network attacks more accurately than traditional DBN networks.

Deep Neural Network (DNN). Traditional machine learning hidden layers usually have less than 5 layers. DNN is a neural network (NN) with multiple hidden layers, usually also called Multi-Layer Perception (or MLP in short). Swarna et al. [89] used Deep Neural Networks (DNN) to identify abnormal cyber attacks in the Internet of Medical Things (IoMT), and used PCA and Grey Wolf Optimization Technique (GWO) to optimize the dimensionality of parameters.

Finally, the indicators, including the accuracy, sensitivity, and specificity, are better than DNN classifiers that do not use dimensionality reduction processing or only use PCA dimensionality reduction. However, the paper puts more emphasis on hybrid dimensionality reduction optimization schemes. Zhou et al. [105] used DNN to perform feature dimensionality reduction to map low-level features to high-level features, and then use traditional machine learning methods for training. Compared with traditional dimensionality reduction methods, they can obtain higher accuracy and reduce training time.

Convolutional Neural Network (CNN). CNN is a special kind of neural network primarily used for image analysis, which can be adapted in the malicious activities detection, including intrusion detection, malware analysis, anomaly detection, *etc*. [106].

Li et al. converted the intrusion detection problem into an image classification problem: convert each part of the data into a matrix and use the CNN model for training [90]. Under the same data set, it is better than other well-known ML or DL methods when the training is under some specific attacks.

Recurrent Neural Network (RNN). RNN has the "memory" ability and can "simulate" the dependency between data. The main point is that there is a pre- and post-dependency relationship between the data, and there is a sequence relationship used in speech analysis, text analysis, time series analysis, *etc*. Common RNNs mainly include GRU and Long Short-Term Memory (LSTM).

McDermott et al. [91] proposed the Bidirectional-LSTM based RNN (BLSTM-RNN) model to detect the four types of attacks used by botnets in IoT. The results indicated that this model is better than the unidirectional LSTM-RNN in *udp* and *dns*. It has a high accuracy rate with Mirai attack, but it performs poorly on ack attack.

GRU is a type of RCNN, like LSTM GRU is proposed to solve the problems of long-term memory and gradients in back propagation. This model can greatly reduce the amount of calculation when the effect is similar to that of LSTM. Chen et al. [92] proposed to use bidirectional gate recurrent unit layer (bi-GRU) on the UNSW and NTHU data sets. The accuracy rate on NTHU reaches 100%, but it does not reach 80% on UNSW. When it detects multiple types of attacks, the performance is not good.

3.4. Data processing vulnerabilities in IIoT

With the development of technology, the attacker's technology has become more and more mature, making traditional defense methods inefficient. Especially for terminal devices in the IIoT with limited computing resources, timely patching of vulnerabilities and preventing attacks are becoming more and more important. Table 3 shows the latest achievements of researchers in this area. Ref. [107] reported that the security and privacy challenges of IIoT devices are endless, and they are increasing every day. From large-scale data breaches to security breaches in billions of chips, to computer system blockades until payment.

Price et al. defined privacy as a set of rules that depend on the application [114]. The author elaborated that the rules for how information flows depend on the entities involved, processes, frequencies, and motivations for accessing the data. There are many applications that need to provide security and protect personal information privacy, such as wearable devices [108], vehicle area network (VANET) [109], healthcare [110], smart home [115].

ML extracts useful features from raw data, and protects privacy by hiding the key information [111]. The ML system has three modules [61]: input, calculation, and output. The work in [61] claims that

Table 3
Existing works on ML based vulnerabilities detection of IIoT data processing.

Ref.	Year	Type of threat	IoT use case	Algo used	Dataset	Acc/Eff
[108]	2018	Information leaking	Wearable device	ML	Purchased equipment	98.5%
[109]	2018	Malicious behavior	Transportation systems	DNN	NSL-KDD data	–
[110]	2016	Privacy protection	Healthcare industry	SVM	Private medical datasets	1.5 s
[111]	2014	Privacy protection	Encryption field	ML	–	–
[61]	2019	Privacy protection	Encryption field	ML	–	–
[112]	2018	Privacy protection	Large scale distributed sys.	SVM	Real-world datasets	–
[113]	2018	Privacy protection	Cloud data collection	DL	MNIST	–

privacy can only be protected when all three modules belong to one entity. Today, billions of IoT devices around the world collect data and cannot maintain a single ownership condition. This question has aroused the interest of researchers, who are committed to proposing updated and improved privacy protection ML algorithms. For example, Zhang et al. raised the issue of the lack of privacy protection in the VANET environment [109], in which connected vehicles may cause privacy issues and malicious vehicles can infer sensitive data from their observed data.

For IIoT terminal devices, *e.g.* Aksu et al. [108] put forward a question about wearable devices, that is, such terminal devices need to be authenticated in the same way as server access. The framework proposed in [108] uses a timing analysis method to analyze Bluetooth traffic: (1) to capture the Bluetooth classic package; (2) to extract features; (3) to use probability distribution to generate fingerprints; and (4) compare the fingerprint stored in (3) with any new input data from the wearable device to identify any unknown wearable device. Selecting the best algorithm, the study claims that its accuracy rate is up to 98.50%.

Data plays a vital role in the training of ML models. For example, we can use a model constructed based on historical IoT data from many factories to make vulnerability predictions for the IIoT in a new environment. However, due to obvious privacy issues, many companies are reluctant to share their data. The researches in [110,112,113] are dedicated to solving these similar data privacy issues. In [110], a new framework, namely eDiag was developed that applies a non-linear kernel SVM to classify medical information while preserving the user privacy and model provider privacy.

Ma et al. mentioned that PKI based data encryption is a widely used privacy protection technology. However, the cost of key management [113]. In order to protect data privacy, a cloud-based DL model that uses multiple keys to achieve user data privacy, namely privacy-preserving DL multiple keys (PDLM), was proposed in [113]. In PDLM, a service provider (SP) sends encrypted user data to the cloud, and the cloud trains the data without knowing the real data. The evaluation results of PDLM shows that, compared with traditional non-private solutions, PDLM successfully protects privacy, but is with less efficient.

3.5. System vulnerabilities in IIoT

In order to adapt to the rapid changes in the IIoT market, many IIoT devices have been deployed. However, the design safety of these devices has not been fully achieved during development. Therefore, malicious attackers use IIoT devices as their targets, and IIoT devices that lack security features are also compromised by attackers, resulting in frequent security incidents.

There are also some vulnerabilities in the IIoT device system, such as the system lacks the ability to *update, firmware verification, rollback, etc.*, or uses insecure components or software libraries, or the device lacks security support, such as update, management, monitoring, response. There are also vulnerabilities in the hardware part, which may be vulnerabilities in the system binary code, vulnerabilities in update maintenance, and so on.

Mahmoud et al. investigated the current status of IoT security in [116] in which they focus on protecting sensor nodes, maintaining trust between devices, defending against man-in-the-middle attacks, and DoS attacks. At the same time, some relevant literature are listed in Table 4, which mainly involves using machine learning methods to identify system vulnerabilities and enhance the security of IIoT systems.

3.5.1. IIoT system security and application security

In terms of the system itself and its scenarios, some vulnerabilities are caused by lack of updates in the system. When software vulnerabilities have been found, developers need to fix the vulnerabilities and release firmware patches. Many users and manufacturers lack of security awareness. Even if latest security patches are released, they are unable to patch the updates, which leads to security vulnerabilities in the devices of the Internet of Things system. For this reason, Vasaka et al. [117] proposed an intrusion prevention system to detect and prevent attacks from external networks. Due to the lack of updates in the security monitoring of some IoT systems, there are some vulnerabilities. These vulnerabilities is proposed in the paper [117], and it is proposed that simple models such as decision trees or random forests can be used for anomaly detection with complex networks such as neural networks.

There are also vulnerabilities caused by the heterogeneity of IoT systems. Liu et al. [118] focused on the heterogeneity of IoT systems aiming at preventing common benchmarks (Such as NSL-KDD dataset) used to test and verify the performance of different IDS.

3.5.2. Security support

With the rapid growth and diversified demands of the market, efforts to quickly provide IIoT services have led to the widespread deployment of IIoT devices with weak security features or insufficient security [119]. These characteristics make IIoT devices a good target for malicious attackers. In many cases, attacks towards IIoT devices have already occurred. Without the use of machine learning, the work in [119] shows that IoT devices lacking security features have also been compromised by attackers, leading to frequent security incidents.

There are also many types of vulnerabilities in security support, such as patch vulnerabilities [120], authentication vulnerabilities, *etc.* In IIoT devices, many of the latest patches, such as binary patches, are not immediately applicable to IIoT devices and need to be adjusted or rolled back. Therefore, techniques to verify that vulnerabilities in a given binary have been patched are critical. In [120], Pengfei et al. proposed a deep learning-based vulnerability and patch presence detection framework for executable binaries. And a cross-platform IoT binary vulnerability search method based on functional semantic learning was proposed by Jian et al. [121].

3.5.3. Hardware vulnerabilities

Hardware vulnerabilities in IIoT system are usually complicated that involves the device design, implementation, or even manufacturing [122–124]. The work in [123,124] reviewed hardware vulnerabilities over IoT devices. *Canedo et al.* studied the use of AI algorithms over gateways to detect anomalies in data sent from edge devices. The AI algorithms can be trained to detect invalid data points [123]. Rashid et al. [124] focused on the attacks and threats of IoT devices in smart city networks connected to sensors connected to large cloud servers. They explored anomaly or intrusion detection schemes using ML algorithms to defend and mitigate IoT network security threats in smart cities.

Table 4
Existing research on vulnerabilities detection of IIoT system using ML algorithms.

Ref.	Year	Type of threat	IoT use case	Algo used	Dataset	Accuracy/efficiency
[64]	2019	System updating	Infrastructure	ML	Anomalous datasets	99.4%
[90]	2020	Heterogeneous systems	IIoT sensors	ML	NSL-KDD dataset	96%
[127]	2019	System attacks	CPS	ML	/	/
[122]	2020	Cyber-attacks	IIoT	DL	CWE_{119}	83.93%
[123]	2016	Abnormal data in device	IIoT gateway	NNs	Collected independently	99%
[124]	2020	Abnormal detection	Smart cities	ML	UNSW-NB15/CICIDS2017	/
[126]	2018	IIoT hardware	Large XOR PUFs	NNs	Random generated	≥70 times

In addition to the vulnerabilities on the device, there are also the vulnerabilities of the system binary code, the vulnerabilities of the update and maintenance, *etc.* Physical Unclonable Function (PUF) is one of promising hardware primitives [125]. The PUF schemes have two main shortcomings in traditional cryptography: (1) the encryption key traditionally stored in digital non-volatile memory, it is susceptible to different types of attacks such as side channel and intrusive physical attacks; (2) limitation of resources, power consumption, and manufacturing costs, electronic devices may not be suitable for using key-based encryption algorithms in certain business scenarios. However, even large-scale XOR PUFs also have vulnerabilities. Ahmad et al. [126] proposed a new machine learning process to attack the exclusive OR arbitration PUF. This means that secure application developers need to identify all potential vulnerabilities in PUF in advance and decide which type of PUF to use in the development process, while unidentified vulnerabilities may pose security risks to IoT devices.

4. Mitigation and exploitation of vulnerabilities

4.1. Vulnerability mitigation strategies in IIoT

4.1.1. Identify vulnerability

In the process of system operation, the ability to identify vulnerabilities is very important. If assets are vulnerable, they are easily to be attacked and expose information. Nowadays, there are many different vulnerability types and identification methods, from technology penetration testing to interview and document analysis, which focus on identifying vulnerabilities in the process. If vulnerabilities are omitted in the identification process, it may lead to incorrect assessment of the security degree, affect the risk calculation, and there will be unknown vulnerabilities in the system. Therefore, continuous and circular risk assessment is necessary. In the IIoT system, it is necessary to conduct file analysis and investigation on physical, hardware, human and natural assets to identify physical, process, natural and human vulnerabilities. Different use cases require different security levels.

Different methods will lead to different information levels. For instance, in the scanning of technical vulnerabilities, the use of automated tools provides less information than penetration testing. In different scenarios, different information will show different sensitivity. In the medical use case, location is a vulnerable vulnerability that affects many different assets, such as hardware, software, and people. We can identify all related vulnerabilities in the IIoT system through a vulnerability list. The level of detail in the vulnerability list can greatly influence the identification process [128].

4.1.2. Security controls

Access control: The proper design and correct use of access control schemes can decrease the risk of vulnerability. Improper access control will cause some attacks from the internet. Some protocols, such as ZigBee and IEC-61850, in IIoT are easily to be attacked by flooding according to Ref. [129]. With the widely used of the SCADA and the weak design of it, it became a favorite target of attackers. To combat the efficient problems and key leakage problems bring by traditional cryptography access control scheme, S. Qi et al. proposed a scheme enabling participants to enforce fine-grained access control policies using a CP-ABE scheme [130].

Procedure control: An orderly and standardized procedures is a prerequisite for the efficient operation of the system, and also a necessary condition for reducing errors. In IIoT, the content of procedure control should include formulating detailed devices design specifications and strictly following them in the production stage. After the devices is put into the production environment, it should be maintained, updated and replaced in time, and the monitoring of devices and detection of security vulnerabilities should be maintained during the production process.

Technical control: Choosing specific security technologies for different devices and environments can also alleviate vulnerabilities. As mentioned in our previous discussion, different technologies, such as some machine learning methods may exhibit varying performance in different scenarios and datasets. Therefore, at the security level of the IIoT, the technical requirements must be controlled strictly to maximize the mitigation of vulnerability hazards.

Authentication is a significant part of protecting personal privacy, using different kinds of mechanisms to identify the legitimate users. Multi-Factor Authentication (MFA) bolsters the authentication processes by combining some strategies together like a password, smartcard, finger print, digital certificate [131], which is suitable for occasions with strict identity verification.

Compliance control: Compliance control is an important part of security control, which can be divided into Policy Control, Governmental Oversight, and Non-Governmental Oversight. Proper compliance, policy and procedure controls can help a company to organize its operations to improve the security lifetime [132]. Compliance control focuses more on macro level control, including the formulation of IIoT rules, legal norms for IIoT security, risk assessment mechanisms, and the development of a series of related standards, such as device design standards, vulnerability levels, and vulnerability lists *etc*.

4.1.3. Deploy IIoT end security

The deploy of IIoT end security should consider the characteristics of end devices. End devices have limited computing capacity, power and storage, are heterogeneous in nature, and are typically deployed in unattended environment. Besides, the sheer number and high mobility nature of IIoT devices exacerbate the difficulty of management of the whole system. K Sha et al. proposed a security service at edge layer to enhance the security of IoT systems base those benefits above, called EdgeSec [133], since the several advantages of edge layer below:

1. Has more resources than Things layer.
2. Close to the IIoT devices.
3. Has more information than edge devices.
4. Stable relationship with IIoT devices.
5. Edge layer can control more personal information to safeguard privacy of individuals.
6. Flexible services needed by IIoT things.
7. Edge computing provide an platform to deploy IIoT end security.

4.1.4. Incident response

Response are needed when incident happened regardless of whether is accidental or deliberate to keep the system operating stably and securely. The industry control systems based on a fundamental core provided by IIoT, were designed on closed networks but now are run on the

cloud over the TCP/IP. The risk of external attack is evident and require more powerful network defence. Destructive events or any interference to them could cause huge financial losses, even threaten people's lives. Therefore, defined procedure and correct methodologies should be taken to responding to those incidents or accidents. Ref. [134] had summarized a SCADA forensic incident response model that contains six main stage: Prepare; Detect; Isolate; Triage; Respond; and Report. A Cook et al. believe that driving architecture improvements and increasing the investment of incident response are necessary [135].

4.1.5. Continuously monitoring

Risk may changes at anytime and anywhere. Changes in IIoT environment may cause the change of known vulnerabilities and threaten. As regular check is not enough, continuously Motoring of cyber security risk is required.

Building a continuous motoring system can support accurate risk assessment, thus can help to make informed decision with a timely response [136]. There are various applications in IIoT. Drones, a typical application of IIoT, requires continuously monitoring and tracking vulnerabilities over every component [137]. In addition to the reliability of network entities, another aspect is the detection of changes in their features when driving the drones connect to others drones or other infrastructure [138]. Besides, Blockchain mechanism can also be a carrier of continuous motoring for further actions and conclusions [139].

In medicine, life motoring devices can detect the basic life characteristics of patients. In IIoT environment, continuously motoring can provide a tools to detect basic operation status of those devices and collect the real-time data to predict the generation of vulnerabilities or deal with problems in time.

4.2. Mitigating exploitation of vulnerability in IIoT

In IIoT, potential vulnerabilities can be categorized into following groups: *memory management API vulnerability, and side channel vulnerability*.

4.2.1. Memory management

Memory management is inevitable for developers, but developers often create a series of vulnerabilities that can be exploited by attackers due to improper use of memory. For IIoT, the memory management problem mainly occurs in the system firmware. Improper memory management during the development of the system firmware may have irreversible consequences.

The Common Weakness Enumeration (CWE) database is a community-developed list of common vulnerabilities in software and hardware weaknesses [140], which includes details to categorize and describe security issues.

Out-of-bounds access (CWE-119) [141] occurs when an out-of-bounds memory object may be accessed. Exploiting such memory vulnerabilities, an attacker can conduct low level attacks, including *code injection, hijacking attack, etc.*

A buffer overflow (CWE-120) [142], is a specific weakness of memory corruption, exploiting CWE-120 an attacker can overwrite the memory buffer of a program and may lead to arbitrary code execution, crashes, or unauthorized access.

A stack-based buffer overflow (CWE-121) [143] is a software vulnerability an attacker can exploit by providing input that exceeds the buffer's capacity.

Generally speaking, the way to cause buffer overflow is not only that the input data is too long to cause the area behind the buffer to be overwritten, but also some out-of-bounds reading and writing can be achieved through format string and numerical vulnerabilities.

The format string vulnerability is caused by the programmer's improper use of format functions such as *printf, sprintf* and other functions. An attacker can obtain the stack address and modify it through this vulnerability.

Raposo et al. [144] used OCSVM and kNN classifier to detect that when IoT device sensors collect temperature and send network, a large amount of data is sent out of the buffer, covering the stack space pointed to by SP, resulting in overflow vulnerability. The authors were able to detect injection faults with lower false positive rate and better recall.

4.2.2. API vulnerability

Application Program Interface, also known as API, refers to library functions or interfaces provided by developers to implement complex operations. Program developers can use API to implement a series of complex functions without knowing the implementation details. API vulnerabilities are getting more and more attention. The source code written by the developers has no vulnerabilities, but if the API has vulnerabilities, it will cause a wide-ranging impact.

In September 2021, Apache Log4j was exposed to a serious and high-risk remote code execution vulnerability [145].

4.2.3. Side channel vulnerability

Side-channel attacks mainly steal information indirectly by exploiting unintended information leakage, also known as side-channel signals, which are inadvertently generated during computation. Typical side channel attacks include: *timing attack, power analysis attack, electromagnetic analysis attack, cache attack, etc.* Side-channel can compromise cryptography systems by extracting secret keys or plaintext data used in encryption scheme.

5. Practical application and future direction

Although the current application of machine learning has made great progress in vulnerability detection technology in the IIoT environment, there are still some issues and challenges that need to be addressed. In this section, we outline the five main concerns faced and future research direction.

Device authentication. For device authentication, traditional IoT technologies have low efficiency and limited application scenarios. And some vulnerabilities are caused by user actions rather than application work. However, traditional technologies do not consider these aspects, and the knowledge of machine learning can be introduced at this time. Data modeling can be based on user actions, employing machine learning models to track user actions and advise users in real-time on the possible security implications of actions taken. Moreover, machine learning can also be combined with traditional technologies to better improve detection efficiency. For authentication in different scenarios, multiple data sets can also be collected, so that traditional technologies can be used in multiple scenarios.

Device standard. In the perception layer of the IIoT, that is, the part close to the physical bottom layer and the physical firmware, once there is a failure, or the function is paralyzed due to its own vulnerability, intentional or unintentional attack, which may cause serious consequences. The scenarios and reasons for adverse consequences are different. Most machine learning algorithms often have limited application scenarios. According to [76], different types of events can trigger different ml algorithms. It not only updates the state vector, but also extracts a set of features — grouped into the feature vector and sends it to the relevant ML algorithm. This idea is a reasonable research direction. In addition, how to make the developed machine learning algorithm self-adaptive and automatically adjust the strategy according to different scenarios and data will be an important task.

The lack of unified standards for IoT equipment or the lack of implementation of minimum IoT security standards is not only a difficult problem in the current IoT system, but also a difficult problem that the IIoT has to face to ensure the safety and long-term interests of equipment [146]. One idea is to start from the hardware, standardize the data source from the sensors and other basic equipment generated by the data, and with a relatively unified data standard, the machine

learning algorithm will have the opportunity to effectively detect vulnerabilities and unsafe data from different devices. However, how such a data standard should be and how to popularize it still need further research and industrial efforts.

Network detection. Because the Internet of Things has the characteristics of wide distribution, large number, and weak physical equipment performance, it is not possible to blindly copy them to build the Internet of Things security system. With the gradual failure of static protection, the new type of network security detection usually adopts active defense. At present, machine learning or deep learning with more hidden layers are often used to build the defense system of the IoT.

Machine learning usually requires a large amount of data for training. If such a huge amount of data cannot be processed on one IoT device, there are many distributed machine learning methods applied to IoT vulnerability detection. More in-depth research, so that IoT devices do not consume a lot of resources for security detection, but use resources to process the application scenarios in which they are located.

In terms of feature extraction of network vulnerabilities, the current data is usually encrypted and disguised. The features extracted based on machine learning largely depend on the similarity of traffic, so we hope to have more advanced methods for feature extraction in the future.

Data processing. In the current IIoT environment, the amount of data generated, transmitted and stored by various devices and the cloud has increased dramatically, while traditional data processing technologies are too inefficient and have poor performance, and some ML algorithms are also inefficient in IIoT. Among the vast amounts of data generated by billions of IIoT devices, different devices also have different data content and formats, making it more challenging to develop standard processing algorithms or techniques.

Due to the diversity and complexity of the IIoT environment, the current data processing technology applications introduced by machine learning are still very limited, either can only be used in a single scenario, or cannot effectively process data streams or massive data. Future research can focus on the adaptive adjustment of the model, and can continuously adjust itself according to the real-time data or new vulnerability data in the IIoT, so as to have a better recognition rate and operating efficiency for different IIoT scenarios. How to design novel machine learning algorithm models to process data in different scenarios and strike a balance between privacy and efficiency will be the focus of future work. In addition, how such models can better deploy resource-constrained terminal equipment and reduce overhead is also a direction worthy of attention.

Static analysis. Due to the low efficiency and low precision of the current traditional technology of the Internet of Things. And different types of codes need to be written for different software types. Machine learning is more accurate than traditional techniques. However, under the massive data of IIoT devices, different static codes are measured in different ways, and there are various static detection methods. A more unified standard is needed to divide static codes in order to better introduce machine learning, and for In the complex IoT environment, the data modeling of static code also needs to be used for different scenarios. Due to the diversity of IIoT device environments and the limited scenarios of machine learning applications, more research is needed in this area.

6. Conclusion and discussion

Cyber attacks are shifting from the widely used IoT to industrial systems by exploiting vulnerabilities in IIoT systems, which involves cyber extortion, nonrecoverable data corruption, hardware corruption, data theft, and more. This work reviews the most recent research results enhancing the security of IIoT, specifically, detailed algorithms were analyzed how machine learning algorithms can improve the security, including security requirements identification, anomaly detection, intrusion detection, behavior analysis, data privacy, continuous improvement, etc. IIoT security is an ongoing process, and the threat landscape is constantly evolving, this work also discussed the security trends and measures, and machine learning models to adopt to new threats and vulnerability.

CRediT authorship contribution statement

Chunchun Ni: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Shan Cang Li:** Data curation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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